

CHAPTER -2

ANALYSIS OF DATA

Introduction

Food analysis usually involves making a number of repeated measurements on the same sample to provide confidence that the analysis was carried out correctly and to obtain a best estimate of the value being measured and a statistical indication of the reliability of the value. A variety of statistical techniques are available that enable us to obtain this information about the laboratory sample from multiple measurements.

Several mathematical treatments are available that provide an idea of how well a particular assay was performed or how well we can reproduce an experiment.

Fortunately, the statistics are not too involved and apply to most analytical determinations. The focus of this chapter is primarily on how to evaluate replicate analyses of the same sample for accuracy and precision. In addition, considerable attention is given to the determination of best line fits for standard curve data. Keep in mind as you read and work through this chapter that there is a vast array of computer software to perform most types of data evaluation and calculations/plots.

Measures of Central Tendency

To increase accuracy and precision, as well as to evaluate these parameters, the analysis of a sample is usually performed (repeated) several times. At least three assays are typically performed, though often the number can be much higher. Because we are not sure which value is closest to the true value, we determine the mean (or average) using all the values obtained and report the results of the **mean**. The mean is designated by the symbol $\bar{x}$ and calculated according to the equation below.

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum x_i}{n} \quad \text{----- (4.1)}$$

Where:

➡ = mean

x_1, x_2 etc = individually measured values (x_i)

n = number of measurements

When we report the mean value, we are indicating that this is the best experimental estimate of the value. We are not saying anything about how accurate or true this value is. Some of the individual values may be closer to the true value, but there is no way to make that determination, so we report only the mean.

Another determination that can be used is the median, which is the midpoint or middle number within a group of numbers. Basically, half of the experimental values will be less than the median and half will be greater. The median is not used often, because the mean is such a superior experimental estimator.

Reliability of Analysis

Accuracy and Precision

One of the most confusing aspects of data analysis for students is grasping the concepts of accuracy and precision. These terms are commonly used interchangeably in society, which only adds to this confusion. If we consider the purpose of the analysis, then these terms become much clearer. If we look at our experiments, we know that the first data obtained are the individual results and a mean value ($\bar{x}$). The next questions should be: “How close were our individual measurements?” and “How close were they to the true value?”

Both questions involve accuracy and precision. Now, let us turn our attention to these terms.

Accuracy refers to how close a particular measure is to the true or correct value.

The problem in determining accuracy is that most of the time we are not sure what the true value is. For certain types of materials, we can purchase known samples from, for example, the National Institute of Standards and Technology and check our assays against these samples. Only then can we have an indication of the accuracy of the testing procedures.

Another approach is to compare our results with those of other labs to determine how well they agree, assuming the other labs are accurate.

Precision: This parameter is a measure of how reproducible or how close replicate measurements become. If repetitive testing yields similar results, then we would say that the precision of that test was good. From a true statistical view, the precision often is called error, when we are actually looking at experimental variation. So, the concepts of precision, error, and variation are closely related.

The difference between precision and accuracy can be illustrated best with Fig. 4-1. Imagine shooting a rifle at a target that represents experimental values. The bull's eye would be the true value and where the bullets hit would represent the individual experimental values. As you can see in Fig. 4-1a, the values can be tightly spaced (good precision) and close to the bull's eye

(good accuracy), or, in some cases, there can be situations with good precision but poor accuracy (Fig. 4-1b). The worst situation, as illustrated in Fig. 4-1d, is when both the accuracy and precision are poor. In this case, because of errors or variation in the determination, interpretation of the results becomes very difficult.

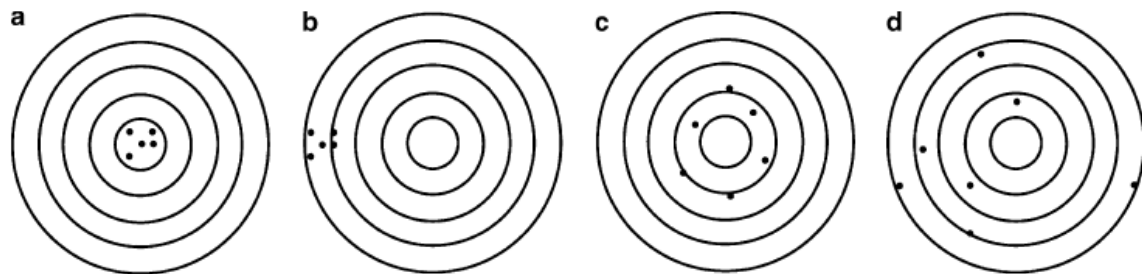


Figure 4.1 Comparison of accuracy and precision: (a) Good accuracy and good precision, (b) Good precision and poor accuracy, (c) Good accuracy and poor precision, and (d) Poor accuracy and poor precision.

Measure of Spread of Data

The spread of the data *is* a measurement of how closely together repeated measurements are to each other. The standard deviation is the most commonly used measure of the spread of experimental measurements. This is determined by assuming that the experimental measurements vary randomly about the mean, so that they can be represented by a normal distribution. The standard deviation *SD* of a set of experimental measurements is given by the following equation:

$$SD = \frac{\sqrt{\sum (x_i - \bar{x})^2}}{n} \quad (4.2)$$

If we have many samples, then the standard deviation is designated by *SD*. It is calculated according to Equation 4.2, assuming all of the food products were evaluated (which would be an infinite amount of assays).

Where:

SD = standard deviation

n = total population of sample

x_i = individual sample values

$\bar{x}$ = true mean

If the number of replicate determinations is small (about 30 or less), which is common with most assays, the n is replaced by the $n-1$ term, and Equation is used. Unless you know otherwise, Equation is always used in calculating the standard deviation of a group of assays.

$$SD = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} \quad \text{----- (4.3)}$$

Depending on which of the equations above is used, the standard deviation may be reported as SDn and SD_{n-1} . (Different brands of software and scientific calculators sometimes use different labels for the keys, so one must be careful.) Table 4-1 shows an example of the determination of standard deviation. The sample results would be reported to average 64.72% moisture with a standard deviation of 0.293.

Table 4.1 Determination of the Standard Deviation of Percent Moisture in Uncooked Hamburger

<i>Measurement</i>	<i>Observed % Moisture</i>	<i>Deviation from the Mean ($x_i - \bar{x}$)</i>	<i>($x_i - \bar{x}$)²</i>
1	64.53	-0.19	0.0361
2	64.45	-0.27	0.0729
3	65.10	+0.38	0.1444
4	64.78	+0.06	0.0036
$\sum x_i = 258.86$		$\sum (x_i - \bar{x})^2 = 0.257$	

$$\bar{x} = \frac{\sum x_i}{n} = \frac{258.86}{4} = 64.72$$

$$SD = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} = \sqrt{\frac{0.257}{3}} = 0.2927$$

Another parameter that is commonly used to provide an indication of the relative spread of the data around the mean is the *coefficient of variation*:

$$CV = \frac{SD}{\bar{x}} \times 100\% \quad \text{----- (4.4)}$$

When evaluating data, several tests are commonly used to give some appreciation of how much the experimental values would vary if we were to repeat the test (indicators of precision). An easy way to look at the variation or scattering is to report the range of the experimental values.

The range is simply the difference between the largest and smallest observation. This measurement is not too useful and thus is seldom used in evaluating data. Probably the best and most commonly used statistical evaluation of the precision of analytical data is the standard deviation. The standard deviation measures the spread of the experimental values and gives a good indication of how close the values are to each other. When evaluating the standard deviation, one has to remember that we are never able to analyze the entire food product. That would be difficult, if not impossible, and very time consuming. Thus, the calculations we use are only estimates of the unknown true value. Once we have a mean and standard deviation, we next must determine how to interpret these numbers. One easy way to get a feel for the standard deviation is to calculate what is called the coefficient of variation (CV), also known as the relative standard deviation.

This calculation is shown below for our example of the moisture determination of uncooked hamburger.

$$\text{Coefficient of Variation (CV)} = \frac{SD}{\bar{x}} \times 100\%$$

$$CV = \frac{0.293}{64.72} \times 100\% = 0.453\%$$

The CV tells us that our standard deviation is only 0.453% as large as the mean. For our example, that number is small, which indicates a high level of precision or reproducibility of the replicates. As a rule, a CV below 5% is considered acceptable, although it depends on the type of analysis.

Values for Z for Checking both Upper and Lower Levels

Degree of certainty (confidence)(%)	Z value
80	1.29
90	1.64
95	1.96
99	2.58
99.9	3.29

The confidence limit (or interval) for our moisture data, assuming a 95% probability, is calculated according to Equation 4.4. Since this calculation is not valid for small numbers, assume we ran 25 samples instead of four.

----- (4.5)

$$\begin{aligned} \text{CI (at 95\%)} &= 64.72 \pm 1.96 \times \frac{0.2927}{\sqrt{25}} \\ &= 64.72 \pm 0.115\% \end{aligned}$$

Because our example had only four values for the moisture levels, the confidence interval should be calculated using statistical *t* tables. In this case, we have to look up the *t* value from Table 4-3 based on the degrees of freedom, which is the sample size minus one ($n - 1$), and the desired level of confidence.

Values of *t* for Various Levels of Probability

Degree of freedom(<i>n</i> -1)	Levels of certainty		
	95%	99%	99.9%
1	12.7	63.7	636
2	4.3	9.93	31.6
3	3.18	5.84	12.9
4	2.78	4.60	8.61
5	2.57	4.03	6.86
6	2.45	3.71	5.96
7	2.36	3.5	5.40
8	2.31	3.56	5.04
9	2.26	3.25	4.78
10	2.23	3.17	4.59

The calculation for our moisture example with four samples (n) and three degrees of freedom ($n-1$) is given below:

$$\begin{aligned} & \text{----- (4.6)} \\ CI(at\ 95\%) &= 64.72 \pm 3.18 \times \frac{0.2927}{\sqrt{4}} \\ &= \underline{64.72} \pm \underline{0.465\ \%} \end{aligned}$$

To interpret this number, we can say that, with 95% confidence, the true mean for our moisture will fall within $64.72 \pm 0.465\%$ or between 65.185 and 64.255%.

The expression $\frac{SD}{\sqrt{n}}$ is often reported as the standard error of the mean. It is then left to the reader to calculate the confidence interval based on the desired level of certainty.

Other quick tests of precision used are the relative deviation from the mean and the relative average deviation from the mean. The relative deviation from the mean is useful when only two replicates have been performed. It is calculated according to Equation, with values below 2% considered acceptable.

$$\text{----- (4.7)}$$

Where:

x_i = individual sample value

$\bar{x}$ = mean

If there are several experimental values, then the relative average deviation from the mean becomes a useful indicator of precision. It is calculated similarly to the relative deviation from the mean, except the average deviation is used instead of the individual deviation. It is calculated according to Equation.

Relative average deviation from the mean:

$$= \frac{\sum |x_i - \bar{x}|}{\frac{n}{\bar{x}}} \times 1000 = \text{parts per thousand} \quad \text{----- (4.8)}$$

Using the moisture values discussed in Table 4-1, the $x_i - \bar{x}$ terms for each determination are $-0.19, -0.27, +0.38, +0.06$. Thus, the calculation becomes:

$$\begin{aligned} \text{Rel. avg. dev.} &= \frac{0.19 + 0.27 + 0.38 + 0.06}{\frac{4}{64.72}} \times 1000 \\ &= \frac{0.225}{64.72} \times 1000 \\ &= 3.47 \text{ parts per thousand} \quad [\end{aligned}$$

Up to now, our discussions of calculations have involved ways to evaluate precision. If the true value is not known, we can calculate only precision. A low degree of precision would make it difficult to predict a realistic value for the sample.

However, we may occasionally have a sample for which we know the true value and can compare our results with the known value. In this case, we can calculate the error for our test, compare it to the known value, and determine the accuracy. One term that can be calculated is the **absolute error**, which is simply the difference between the experimental value and the true value.

$$\text{Absolute Error} = E_{\text{abs}} = x - T \quad \text{----- (4.9)}$$

Where:

x = experimentally determined value

T = true value

The absolute error term can have either a positive or a negative value. If the experimentally determined value is from several replicates, then the mean ($\bar{x}$) would be substituted for the x term. This is not a good test for error, because the value is not related to the magnitude of the true value. A more useful measurement of error is relative error.

$$\text{Relative error} = E_{\text{rel}} = \frac{E_{\text{abs}}}{T} = \frac{x - T}{T}$$

$$\text{-----} (4.10)$$

The results are reported as a negative or positive value, which represents a fraction of the true value. If desired, the relative error can be expressed as percent relative error by multiplying by 100%. Then the relationship becomes the following, where x can be either an individual determination or the mean ($\bar{x}$) of several determinations.

$$\%E_{\text{rel}} = \frac{E_{\text{abs}}}{T} \times 100\% = \frac{x - T}{T} \times 100\% \text{-----} (4.11)$$

Using the data for the percent moisture of uncooked hamburger, suppose the true value of the sample is 65.05%.

The percent relative error is calculated using our mean value of 64.72% and Equation.

$$\begin{aligned} \%E_{\text{rel}} &= \frac{\bar{x} - T}{T} \times 100\% \\ &= \frac{64.72 - 65.05}{65.05} \times 100\% = -0.507\% \end{aligned}$$

Note that we keep the negative value, which indicates the direction of our error, that is, our results were 0.507% lower than the true value.

1. Sources of Error

There are three common sources of error in any analytical technique:

Personal Errors (Blunders): These occur when the analytical test is not carried out correctly: the wrong chemical reagent or equipment might have been used; some of the sample may have been spilt; a volume or mass may have been recorded incorrectly; etc. It is partly for this reason that analytical measurements should be repeated a number of times using freshly prepared laboratory samples. Blunders are usually easy to identify and can be eliminated by carrying out the analytical method again more carefully.

Random Errors: These produce data that vary in a non-reproducible fashion from one measurement to the next *e.g.*, instrumental noise. This type of error determines the standard

deviation of a measurement. There may be a number of different sources of random error and these are accumulative (see Propagation of Errors).

Systematic Errors: A systematic error produces results that consistently deviate from the true answer in some systematic way, *e.g.*, measurements may always be 10% too high. This type of error would occur if the volume of a pipette was different from the stipulated value. For example, a nominally 100 cm³ pipette may always deliver 101 cm³ instead of the correct value.

To make accurate and precise measurements it is important when designing and setting up an analytical procedure to identify the various sources of error and to minimize their effects.

Often, one particular step will be the largest source of error, and the best improvement in accuracy or precision can be achieved by minimizing the error in this step.

Curve fitting is a generic term used to describe the relationship and evaluation between two variables. Most scientific fields use curve fitting procedures to evaluate the relationship of two variables. Thus, curve fitting or curvilinear analysis of data is a vast area as evidenced by the volumes of material describing these procedures. In analytical determinations, we are usually concerned with only a small segment of curvilinear analysis, the standard curve or regression line. A **standard curve** or **calibration curve** is used to determine unknown concentrations based on a method that gives some type of measurable response that is proportional to a known amount of standard.

It typically involves making a group of known standards in increasing concentration and then recording the particular measured analytical parameter (*e.g.*, absorbance, area of a chromatography peak, etc.).

What results when we graph the paired *x* and *y* values is a scatter plot of points that can be joined together to form a straight line relating concentration to observed response. Once we know how the observed values change with concentration, it is fairly easy to estimate the concentration of an unknown by interpolation from the standard curve.

As you read through the next three sections, keep in mind that not all correlations of observed values to standard concentrations are linear (but most are).

There are many examples of nonlinear curves, such as antibody binding, toxicity evaluations, and exponential growth and decay. Fortunately, with the vast array of computer software available today, it is relatively easy to analyze any group of data.

With modern day instruments, the error can be very small. Although arguments can be made for making the y -axis data concentration, for all practical purposes the end result is essentially the same. Unless there are some unusual data, the *concentration should be associated with the x -axis and the measured values with the y -axis.*

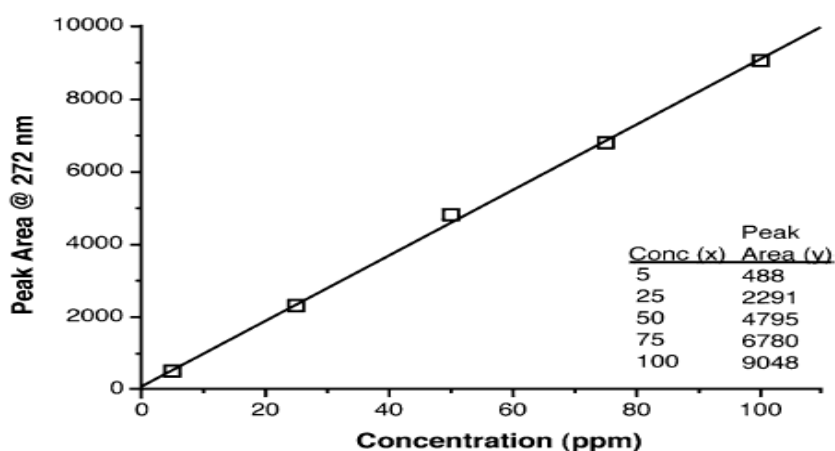


Figure 4.3 A typical standard curve plot showing the data points and the generated best fit line.

The data used to plot the curve are presented on the graph.

Figure 4-3 illustrates a typical standard curve used in the determination of caffeine in various foods. Caffeine is analyzed readily in foods by using HPLC coupled with an ultraviolet detector set at 272 nm. The area under the caffeine peak at 272nm is directly proportional to the concentration. When an unknown sample (e.g., coffee) is run on the HPLC, a peak area is obtained that can be related back to the sample using the standard curve.

The plot in Fig. 4-3 shows all the data points and a straight line that appears to pass through most of the points. The line almost passes through the origin, which makes sense because zero concentration should produce no signal at 272 nm. However, the line is not perfectly straight (and never is) and does not quite pass through the origin.

To determine the caffeine concentration in a sample that gave an area of say 4000, we could extrapolate to the line and then draw a line down to the x -axis. Following a line to the x -axis (concentration), we can estimate the solution to be at about 42–43 ppm of caffeine.

We can mathematically determine the best fit of the line by using **linear regression**. Keep in mind the equation for a straight line, which is $y = ax + b$, where a is the slope and b is the y-intercept. To determine the slope and y-intercept, the regression equations shown below are used. We determine a and b and thus, for any value of y (measured), we can determine the concentration (x).

$$\begin{aligned} \text{Slope } a &= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \\ \text{y-intercept } b &= \bar{y} - a\bar{x} \end{aligned} \quad (4.13)$$

where:

x_i and y_i = individual values

$\bar{x}$ and $\bar{y}$ = means of the individual values

Correlation Coefficient

In observing any type of correlation, including linear ones, questions always surface concerning how to draw the line through the data points and how well the data fit to the straight line. The first thing that should be done with any group of data is to plot it to see if the points fit a straight line. By just looking at the plotted data, it is fairly easy to make a judgment on the linearity of the line. We also can pick out regions on the line where a linear relationship does not exist. The figures below illustrate differences in standard curves;

Fig. 4-4a shows a good correlation of the data and Fig. 4-4b shows a poor correlation. In both cases, we can draw a straight line through the data points. Both curves yield the same straight line, but the precision is poorer for the latter.

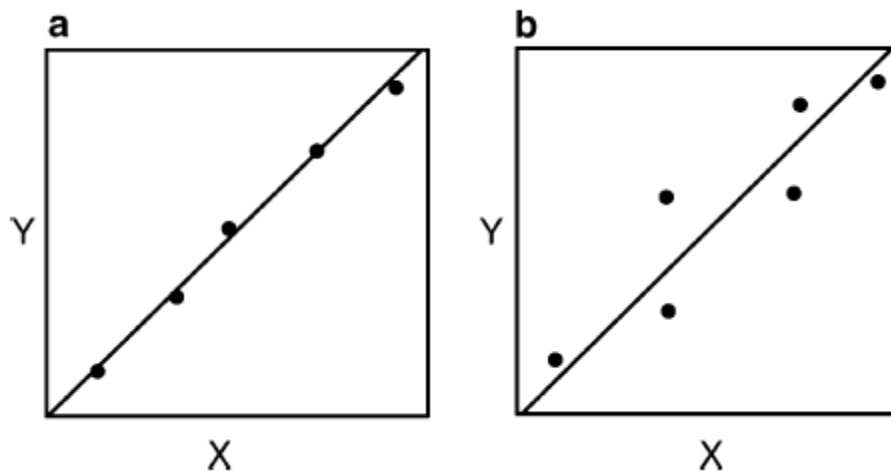


Figure 4.4 Examples of standard curves showing the relationship between the x and y variables when there is (a) a high amount of correlation and (b) a lower amount of correlation. Both lines have the same equation.

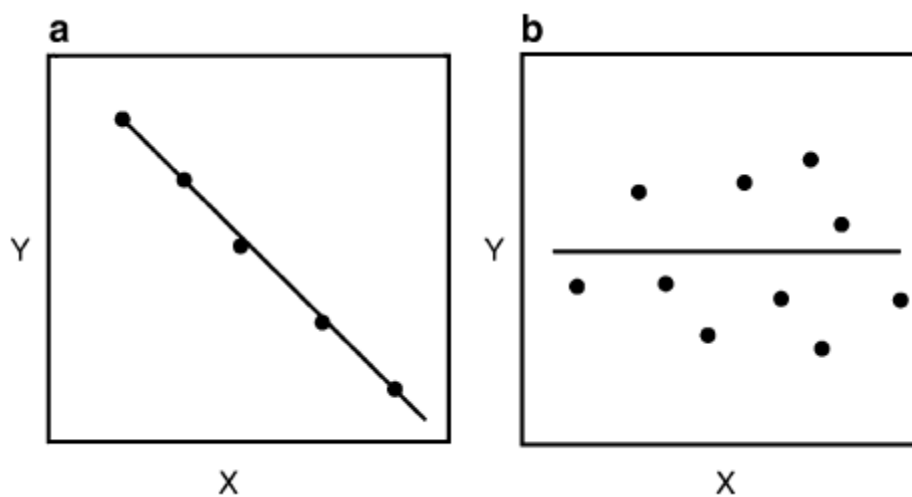


Figure 4.5 Examples of standard curves showing the relationship between the x and y variables when there is (a) a high amount of negative correlation and (b) no correlation between x and y values.

There are other possibilities when working with standard curves. Figure 4-5a shows a good correlation between x and y , but in the negative direction, and Fig. 4-5b illustrates data that have no correlation at all.

The **correlation coefficient** defines how well the data fit to a straight line. For a standard curve, the ideal situation would be that all data points lie perfectly on a straight line. However, this is never the case, because errors are introduced in making standards and measuring the physical values (observations).

For standard curves, we want the value of r as close to $+1.0000$ or -1.000 as possible, because this value is a perfect correlation (perfect straight line). Generally, in analytical work, the r should be 0.9970 or better. (This does not apply to biological studies).

The **coefficient of determination** (r^2) is used often because it gives a better perception of the straight line even though it does not indicate the direction of the correlation. The r^2 for the example presented above is 0.99886 , which represents the proportion of the variance of absorbance (y) that can be attributed to its linear regression on concentration (x). This means that about 0.114% of the straight line variation ($1.0000 - 0.99886 = 0.00114 \times 100\% = 0.114\%$) does not vary with changes in x and y and thus is due to indeterminate variation. A small amount of variation is expected normally.

Reporting Results

In dealing with experimental results, we are always confronted with reporting data in a way that indicates the sensitivity and precision of the assay. Ideally, we do not want to overstate or understate the sensitivity of the assay, and thus we strive to report a meaningful value, be it a mean, standard deviation, or some other number. The next three sections discuss how we can evaluate experimental values so as to be precise when reporting results.

Rejecting Data

Inevitably, during the course of working with experimental data we will come across a value that does not match the others. Can you reject that value and thus not use it in calculating the final reported results? The answer is “sometimes,” but only after careful consideration. If you are routinely rejecting data to help make your assay look better, then you are misrepresenting the results and the precision of the assay.

If the bad value resulted from an identifiable mistake in that particular test, then it is probably safe to drop the value. Again, caution is advised because you may be rejecting a value that is closer to the true value than some of the other values.

Consistently poor accuracy or precision indicates that an improper technique or incorrect reagent was used or that the test was not very good. It is best to make changes in the procedure or change methods rather than try to figure out ways to eliminate undesirable values.

There are several tests for rejecting an aberrant value. One of these tests, the **Q-Test**, is commonly used. In this test, a **Q-value** is calculated as shown below and compared to values in a table. If the calculated value is larger than the table value, then the questionable measurement can be rejected at the 90% confidence level.

$$Q\text{-value} = \frac{x_2 - x_1}{W} \quad (4.15)$$

where:

x_1 = questionable value

x_2 = next closest value to x_1

W = total spread of all values, obtained by subtracting the lowest value from the highest value

Table 4-4 provides the rejection Q -values for a 90% confidence level.

Q-Values for the Rejection of Results

Number of Observations	Q of Rejection (90% leve)
3	0.94
4	0.76
5	0.64
6	0.56
7	0.51
8	0.47
9	0.44
10	0.41

The example below shows how the test is used for the moisture level of uncooked hamburger for which four replicates were performed giving values of 64.53, 64.45, 64.78, and 55.31. The 55.31 value looks as if it is too low compared to the other results. Can that value be rejected? For our

example, x_1 is the questionable value (55.31) and x_2 is the closest neighbour to x_1 (which is 64.45). The spread (W) is the high value minus the low measurement, which is $64.78 - 55.31$.

$$Q\text{-value} = \frac{64.45 - 55.31}{64.78 - 55.31} = \frac{9.14}{9.47} = 0.97$$

From Table 4-4, we see that the calculated Q -value must be greater than 0.76 to reject the data. Thus, we make the decision to reject the 55.31% moisture value and do not use it in calculating the mean.

Summary

This chapter focuses on statistical methods to measure data variability, precision, etc. and on basic mathematical treatment that can be used in evaluating a group of data. For example, it should be almost second nature to determine a mean, standard deviation, and CV when evaluating replicate analyses of an individual sample. In evaluating linear standard curves, best line fits should always be determined along with the indicators of the degree of linearity (correlation coefficient or coefficient of determination). Fortunately, most computer spreadsheet and graphics software will readily perform the calculations for you. Guidelines are available to enable one to report analytical results in a way that tells something about the sensitivity and confidence of a particular test. A section is included which describes sensitivity and LOD as related to various analytical methods and regulatory agency policies.

Additional information includes the proper use of significant figures, rules for rounding off numbers and use of the Q -test to reject grossly aberrant individual values.